

## HAILUOTO MARJANIEMI, Finland

WMO: 028730

Lat: 65.0400N

Lon: 24.5615E

Elev: 24

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -11.1      | -5.4 | -15.5                         | 2.4 | -11.0 | -9.0 | 3.4 | -5.2 | 42.6                  | 29.3 | 39.3 | 29.4 | 8.2                   | 130  | 0.872 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.3                    | 74.5            | 65.9 | 71.0 | 64.0 | 68.1 | 62.6 | 67.6                | 72.5 | 65.3 | 69.6 | 63.3 | 67.4 | 10.7                 | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.7                            | 95.0 | 70.1 | 63.6 | 88.2 | 68.0 | 61.5 | 81.6 | 65.8 | 32.0          | 72.3 | 30.3 | 69.5 | 28.8 | 67.9 | 73.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 38.4          | 34.2 | 30.5 | -13.4                      | 79.2 | 7.9                | 5.1 | -19.1                                              | 82.9 | -23.7      | 85.9 | -28.1      | 88.8 | -33.9      | 92.5 |    |    |
| (5)               |      |      | -13.7                      | 69.2 | 7.8                | 3.6 | -19.3                                              | 71.8 | -23.9      | 73.9 | -28.3      | 76.0 | -34.0      | 78.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 38.7   | 19.3  | 17.8  | 22.6  | 32.4  | 44.5  | 53.9  | 61.5  | 59.4  | 51.5  | 40.7  | 32.7  | 27.0 |
|                                                                                  | DBStd       | 16.80  | 11.43 | 12.18 | 9.29  | 5.53  | 7.46  | 5.66  | 5.37  | 4.84  | 4.94  | 5.95  | 6.95  | 9.47 |
|                                                                                  | HDD50       | 5021   | 951   | 901   | 850   | 528   | 205   | 22    | 0     | 1     | 41    | 291   | 518   | 713  |
|                                                                                  | HDD65       | 9629   | 1416  | 1321  | 1315  | 978   | 637   | 335   | 138   | 184   | 405   | 754   | 968   | 1178 |
|                                                                                  | CDD50       | 913    | 0     | 0     | 0     | 0     | 34    | 141   | 357   | 293   | 86    | 2     | 0     | 0    |
|                                                                                  | CDD65       | 46     | 0     | 0     | 0     | 0     | 2     | 4     | 30    | 10    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 136    | 0     | 0     | 0     | 0     | 10    | 14    | 88    | 24    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 22     | 0     | 0     | 0     | 0     | 1     | 2     | 17    | 2     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 15.0   | 15.2  | 14.7  | 15.4  | 13.8  | 13.8  | 13.7  | 12.4  | 13.7  | 15.8  | 17.0  | 17.1  | 17.0 |
| Precipitation                                                                    | PrecAvg     | 20.5   | 1.5   | 1.3   | 1.2   | 1.1   | 1.5   | 1.7   | 2.5   | 2.2   | 1.9   | 2.0   | 1.9   | 1.7  |
|                                                                                  | PrecMax     | 27.3   | 2.9   | 2.7   | 3.2   | 3.1   | 3.3   | 4.0   | 4.7   | 4.6   | 3.5   | 3.8   | 3.3   | 3.4  |
|                                                                                  | PrecMin     | 14.6   | 0.6   | 0.6   | 0.2   | 0.3   | 0.2   | 0.1   | 0.1   | 0.2   | 0.4   | 0.3   | 0.4   | 0.5  |
|                                                                                  | PrecStd     | 3.3    | 0.7   | 0.7   | 0.7   | 0.6   | 0.7   | 1.1   | 1.2   | 1.0   | 0.9   | 0.8   | 0.7   | 0.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 36.6  | 37.2  | 39.0  | 54.1  | 73.6  | 75.5  | 82.3  | 76.9  | 66.8  | 53.8  | 47.3  | 40.5 |
|                                                                                  |             | MCWB   | 35.1  | 35.2  | 35.3  | 45.6  | 58.8  | 62.7  | 69.0  | 66.7  | 59.8  | 51.3  | 45.1  | 39.1 |
|                                                                                  | 2%          | DB     | 35.1  | 35.5  | 36.7  | 45.9  | 65.2  | 69.2  | 75.6  | 72.1  | 61.9  | 51.9  | 45.5  | 39.1 |
|                                                                                  |             | MCWB   | 34.2  | 34.1  | 34.4  | 39.9  | 54.7  | 60.4  | 67.7  | 64.7  | 57.5  | 49.9  | 44.2  | 37.6 |
|                                                                                  | 5%          | DB     | 33.9  | 34.0  | 35.3  | 41.3  | 59.2  | 64.8  | 72.4  | 68.8  | 59.6  | 50.0  | 43.2  | 37.8 |
|                                                                                  |             | MCWB   | 33.2  | 33.1  | 33.6  | 37.1  | 52.1  | 58.3  | 65.6  | 63.1  | 56.4  | 48.2  | 42.1  | 36.4 |
|                                                                                  | 10%         | DB     | 32.8  | 32.5  | 33.9  | 38.5  | 54.0  | 61.8  | 69.6  | 66.4  | 57.9  | 48.4  | 41.3  | 36.4 |
|                                                                                  |             | MCWB   | 32.1  | 31.8  | 32.7  | 35.7  | 48.3  | 56.4  | 64.2  | 61.8  | 55.2  | 46.6  | 40.2  | 35.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.8  | 35.8  | 36.5  | 46.7  | 60.0  | 64.8  | 71.5  | 69.3  | 60.6  | 52.2  | 45.9  | 39.7 |
|                                                                                  |             | MCDB   | 36.1  | 36.6  | 38.5  | 50.8  | 70.4  | 71.1  | 75.6  | 74.6  | 64.0  | 53.3  | 46.5  | 40.0 |
|                                                                                  | 2%          | WB     | 34.5  | 34.5  | 35.0  | 40.6  | 55.8  | 62.0  | 69.0  | 66.3  | 58.4  | 50.3  | 44.3  | 38.1 |
|                                                                                  |             | MCDB   | 34.9  | 35.2  | 36.1  | 45.2  | 64.3  | 67.2  | 73.8  | 70.3  | 61.1  | 51.5  | 45.4  | 38.7 |
|                                                                                  | 5%          | WB     | 33.3  | 33.3  | 34.0  | 37.9  | 52.6  | 59.4  | 66.7  | 64.4  | 57.1  | 48.6  | 42.4  | 36.6 |
|                                                                                  |             | MCDB   | 33.8  | 33.9  | 35.0  | 40.3  | 57.8  | 63.7  | 71.3  | 67.5  | 59.1  | 49.8  | 43.2  | 37.5 |
|                                                                                  | 10%         | WB     | 32.3  | 32.1  | 32.9  | 36.1  | 49.4  | 57.3  | 64.7  | 62.4  | 55.8  | 46.9  | 40.4  | 35.4 |
|                                                                                  |             | MCDB   | 32.8  | 32.7  | 33.9  | 37.9  | 53.4  | 61.1  | 69.3  | 65.7  | 57.7  | 48.2  | 41.1  | 36.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.2   | 9.9   | 10.9  | 9.1   | 9.5   | 8.3   | 8.3   | 8.1   | 6.7   | 5.8   | 5.9   | 6.9  |
|                                                                                  |             | MCDBR  | 6.4   | 7.0   | 7.9   | 12.9  | 17.4  | 14.2  | 13.3  | 12.4  | 9.2   | 5.9   | 5.7   | 5.5  |
|                                                                                  |             | MCWBR  | 6.4   | 6.6   | 6.6   | 9.0   | 10.0  | 8.2   | 7.6   | 7.6   | 6.1   | 5.2   | 5.6   | 5.1  |
|                                                                                  | 5% WB       | MCDBR  | 6.6   | 6.9   | 7.1   | 11.5  | 15.3  | 11.8  | 11.4  | 9.9   | 8.4   | 5.6   | 5.8   | 5.5  |
|                                                                                  |             | MCWBR  | 6.6   | 6.5   | 6.2   | 8.4   | 9.9   | 7.8   | 7.4   | 7.3   | 6.1   | 5.1   | 5.8   | 5.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.209  | 0.276 | 0.304 | 0.341 | 0.346 | 0.350 | 0.369 | 0.354 | 0.331 | 0.308 | 0.343 | 0.353 |      |
|                                                                                  | taud        | 1.950  | 2.092 | 2.133 | 2.173 | 2.407 | 2.488 | 2.455 | 2.489 | 2.529 | 2.354 | 2.545 | 2.589 |      |
|                                                                                  | Ebn at Noon | 93     | 189   | 241   | 260   | 273   | 275   | 265   | 258   | 240   | 183   | 85    | 35    |      |
|                                                                                  | Edh at Noon | 7      | 19    | 29    | 36    | 31    | 30    | 30    | 26    | 21    | 15    | 7     | 4     |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 33     | 195   | 663   | 1202  | 1575  | 1765  | 1765  | 1629  | 1205  | 659   | 239   | 49    | 10   |
|                                                                                  | RadStd      | 3      | 38    | 67    | 95    | 148   | 148   | 140   | 146   | 113   | 67    | 39    | 8     | 1    |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (42 neighbors) |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page